NORWEGIAN PETROLEUM DIRECTORATE

WELL COMPLETION & GEOLOGICAL OPERATIONS REPORT

WELL IDENTIFIER	: 6407/9-2
LICENCE	: PL093
FIELD / PROSPECT	: Draugen
OPERATOR	: A/S Norske Shell
TARGET FORMATION	: Garn Fm.
REPORT DATE	: 04/1985
DRILLING RIG	: Ross Isle (Semi-submersible)
PRIMARY OBJECTIVE	: Appraisal of hydrocarbon-bearing Middle Jurassic reservoirs
TOTAL DEPTH MEASURED	: 4527.5 m RKB
STATUS AT COMPLETION	: Suspended / P&A as Oil and Gas Discovery

DOCUMENT STRUCTURE & ENCLOSURES:

SECTION 1 : Administrative Information & Rig Positioning

SECTION 2 : Daily Drilling Operations & Mud Engineering Log

SECTION 3 : Lithological Descriptions & Core Analysis Summary

SECTION 4 : Wireline Logging, Test Summaries & Composite Log

WELL 6407/9-2 - CASING & MUD LOG RECORD

HOLE SIZE	CASING O.D.	SHOE DEPTH (m)	MUD TYPE	MUD WEIGHT (s.g.)
36"	30"	220 m	Seawater/Bentonite	1.08
26"	20"	750 m	Gel/Lignosulfonate	1.15
17 1/2"	13 3/8"	1980 m	KCl/Polymer	1.25
12 1/4"	9 5/8"	3540 m	KCl/Polymer	1.32
8 1/2"	7" Liner	4523 m (TD)	Oil-based Low-Tox	1.42

BIT RECORD AND PENETRATION SUMMARY:

Bit #4: 12 1/4" Smith F3 - Drilled 480m in 32 hrs (ROP = 15 m/hr)

Bit #5: 8 1/2" Hughes ATM-11 - Cored interval 3930m to 4050m using Christensen 60ft barrel.

Bit #6: 8 1/2" Security H87F - Drilled to Total Depth at 4523m RKB.

WELL 6407/9-2 - GEOLOGICAL SUMMARY

1.0 STRATIGRAPHIC SUMMARY

The well was spudded into Pleistocene/Holocene seabed sediments and penetrated a normal Cenozoic and Mesozoic succession. The main reservoir target, the Garn Fm., was encountered at expected regional depths, exhibiting high net-to-gross sand ratios.

2.0 HYDROCARBON SHOWS & CORE RECOVERY OVERVIEW

Continuous coring was initiated in the upper Garn Fm. section. Core cut recovery was exceptionally high (>90% average), yielding high-quality unweathered plug samples. Visual examination confirmed strong live hydrocarbon shows, yellow fluorescence, and fast cut-fluorescence in the porous sandstone intervals.

3.0 RESERVOIR LITHOLOGY NOTES

Sandstones are light grey to yellowish-brown, fine to coarse-grained, subangular to subrounded, moderately to well sorted, with subordinate calcite and kaolinite cementation. Interbedded siltstones and claystones are dark grey, carbonaceous, and moderately micaceous.

Detailed tabular coring intervals, cut lengths, and recovery statistics are tabulated on the following Enclosure (Core Log Summary).

CORE LOG

SCALE : 1 : 200

WELL : 6407/9-2 **DATE : 04/1985**

COUNTRY	Norway
LICENCE	PL093
FIELD	Draugen
OWNERS	A/S Norske Shell, Statoil
COORDINATES	GEOGRAPHICAL N64°56'39.65" E06°22'58.13"
ELEVATION KB	25m
WATER DEPTH	240m
TOTAL DEPTH (DRILLER)	2230m
TOTAL DEPTH (LOGGER)	2234.5m
FORMATION AT TD	Are fm.
LOGGERS - DRILLERS	Cores 1-10: 0.5m
GEOLOGISTS	T. Breyer, J.P. Bocketto
COMMENTS	Core recovery verified against gamma log readings.

GROUP/FORMATION: Garn Fm.
AGE: Mellom Jura

OTHER CORES IN WELL

NO	GROUP / FORMATION

CORES DESCRIBED IN THIS LOG

NO	INTERVAL (m)	RECOVERY (%)
1	1630.0 - 1639.7	100
2	1647.7 - 1673.9	100
3	1676.9 - 1702.1	94.2
4	1702.1 - 1727.2	73.5
5	1727.2 - 1744.8	68
6	1744.8 - 1756.3	91.5
7	1756.3 - 1761.5	68
8	1764.5 - 1772.5	100
9	1774.0 - 1789.1	96.5
10	1797.1 - 1820.9	73.5

